ARC-AGI Without Gradients

An explicit-ledger agent learns world rules from raw experience, executes them as plans — and publishes its refusals

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ARC Prize 2026 — Paper Award submission. Documents the approach behind the author's Kaggle submission (submission ID bound at upload).
All code, run records, and pre-registered gates are open under a permissive license.

Abstract. ARC-AGI is usually attacked with more compute: bigger models, longer search, denser pretraining. We report the opposite experiment. **zAgent** is a deterministic cognitive agent whose mind keeps an explicit ledger — every belief is evidence with a frequency and a confidence, revised only by lived outcomes; every prediction is registered before its outcome exists. No gradient, no prompt, no pretrained weights touch the reasoning. Run against ARC-AGI in two surfaces, the system does two things the field rarely publishes: it **executes what it has proven** — replaying recorded ARC-AGI-3 experience into a 61.93 local aggregate (132 levels, 11 full game wins, 7,740 actions, from a 0.0 baseline), and planning a level clear in 7 actions against a 17-action human baseline using a world model it absorbed from another system's public recordings — and it **refuses what it cannot prove**: its pre-registered gates produced a documented negative on the ARC-AGI-1 puzzle venue (1,698 mined rules, quality bar unmet, root cause named), a negative that later became a stamp-time check now enforced before any campaign runs. The contribution is not a leaderboard score. It is a demonstration that the machinery of reasoning — persistent epistemic state, knowledge separated from process, predictions registered ahead of their outcomes — is buildable today, scores on ARC-AGI-3 without a single learned weight, and knows where its own evidence ends.

1 The challenge we accepted

ARC-AGI-3 changed the question. The benchmark is no longer a static exam of grids but an interactive world: the agent acts, the world answers, and the rules are never stated. That is exactly the regime most current methods are weakest in, because their knowledge has no address — a language model that solves a level cannot be *wrong about a rule* in any way that persists, so there is nothing for the world to correct.

We take the position that the bottleneck is not inference strength but **epistemic bookkeeping**: a system that keeps an open ledger of hypotheses, confidence, and evidence — and registers its predictions before acting on them — can learn rules from raw experience, can be audited belief by belief, and can say *no* to its own hunches. This paper measures that position on ARC-AGI rather than arguing it.

Claims, stated in advance:

1. A gradient-free, prompt-free ledger agent can absorb another system's recorded ARC-AGI-3 experience, calibrate its bookkeeping against that artifact's published measurements, and execute the learned world models as plans — scoring where the unaided recordings score zero.
2. The same machinery, pointed at ARC-AGI-1 puzzles, tells the truth about itself: its gates produce an audited negative, with the root cause located, rather than a narrated success.

3. Both results reproduce byte-for-byte. The agent is deterministic: same script, byte-identical transcript, checksummed forks, gates registered before compute.

2 The agent in one page

The architecture has two organs and a contract between them.

The mind (zMind) is an explicit-ledger reasoner (NAL-style non-axiomatic inference, event-window temporal mining). Every claim is a ledger entry — a frequency (how often it has held) and a confidence (how much disjoint evidence stands behind it). Nothing is exempt from revision; belief moves only when the world co-signs. A sliding event window composes co-occurring percepts, operations, and outcomes into predictive chains (`((&/, ^forward, +d) =/> [d_near]`); mined implications are registered as predictions *before* being acted on, and the world's answer grades them. There is no policy network: what acts is a decision route over the desire ledger, and every candidate act must clear a fixed expectation threshold (0.51).

The body is a separate component with one job: at sparse moments, adjust what the mind *wants* — never what it *does*. An emission gate rejects any adviser output that names an action.

The contract, enforced in code: say what to want, never how to act; speak rarely; fail silently and be scored for it.

Three properties do the work for ARC-AGI:

- **The ledger is portable.** A world model is a set of ledger entries. Rules mined in one context — or absorbed from an external artifact — are executable as plans by the same decision route, with no retraining.
- **Falsification is a first-class event.** A prediction that the world contradicts revises the implication that produced it. The transcript *is* the reasoning, not a story told afterward.
- **The agent is deterministic.** Seeded, single-threaded, no wall clock. Every number in this paper is re-derivable from a published run directory.

3 Surface one: ARC-AGI-3 from someone else's experience

3.1 The artifact as corpus

The strongest current line of public ARC-AGI-3 work ships its recordings: 25 games, 24,604 transitions, 183 levels (the source paper's totals; we verified them against the files). We treated these recordings the way the mind treats any experience: as an evidence stream to be mined, calibrated, and — where the evidence supports it — executed.

Calibration before use. Before trusting the artifact, we replayed its own measurement machinery against our ledger: all 19 final concept books settle cleanly on their own runs; all 1,624 recorded falsifications are absorbed by current versions at n=25; the artifact's merge-audit rejection-superset reproduces. Only then did we let its world models speak.

A measured disagreement. The artifact's action gate refuses any move while any concept is false. We measured that design against the corpus and deliberately did not copy it: the recorded falsifications are overwhelmingly *encoder-class* — failures of the eyes, not the world — so an action gate halts winnable games on perception bugs. Our round gate refuses between rounds, on evidence, instead of inside them, on fear.

3.2 Results

Table 1. ARC-AGI-3 measurements: replay and planning over an absorbed corpus

Measurement	Result
Verified winning routes replayed under hash anchoring	**0.0 → 61.93** local aggregate
Levels cleared / full game wins	132 levels, 11 wins
Total actions spent	7,740 (pre-registered accounting)
Planning over an absorbed position-permutation model	opening level cleared in **7 actions** vs 17-action human baseline, deal-independent

Two details matter more than the totals.

The gains were diagnosed, not tuned. The largest single improvement came from a defect the ledger made visible: five of one game's nine levels share a start frame, and an order-first pickup kept re-picking completed levels until a chain cursor replaced it. Attribution was a query, not an archaeology.

No route knowledge leaked into the planner. The 7-action clear used only the artifact's own `move_map` world model executed as rules; no recorded route was consulted.

3.3 The honest negative at maximum evidence

We then ran the experiment the architecture is supposed to make safe: give the mind its best possible rules and let it chase a clear on its own. The 5,000-action clear chase — the largest exploration run of the program, 97% of actions frontier-directed, zero random fallback, reflection authoring at all 125 boundaries, the best mined rule set on record — produced **no level clear**. The post-mortem is exact: zero mined chains head the goal, because the goal event has never been observed; the observable-state frontier saturates at thirteen states. Seven thousand accumulated live actions across every posture confirm the boundary: **no rule lever helps until exploration reaches the goal at all.** One observed goal event would fund every authored hop at once; the machinery to measure everything after that moment is built, gated, and waiting.

We report this as a result, not an apology. A system that cannot locate its own evidence boundary will find it in production instead.

4 Surface two: the ARC-AGI-1 puzzle venue — a documented negative

The puzzle venue was stamped with its scope *computed*, not asserted: the operator monoid's witness closure over the 400 ARC-AGI-1 training tasks yields **9 in-model operators** (7 dihedral, 2 crop) — a number the gauge inherits, replacing a claim with a measurement.

The arc, in three rounds:

1. **Mechanism round: PASS.** The full causal chain (compose → believe → abduce → fire) proves out on authored evidence, with the zero-seed ablation silent.
2. **Real-stream round: FAIL, documented.** On live puzzle streams the miner stays alive (1,698 rules) but mines step-boundary compounds at the +24-cycle seam — max confidence 0.4475, prediction resolutions 0.708 per episode against a pre-registered bar of 1.0. The single rule that crossed the confidence bar was born exactly where an operation fired and matched. Round three did not launch; the levers were named in the venue record.
3. **Reflection round: the named levers work.** The reflection stage (a bench-side optimizer over the mind's own rule inventory — six deterministic transforms, gated pairwise) took the round-3 cap clauses to PASS for the first time: per-arm prediction resolutions of 1.17–1.25 against the 1.0 bar, with the control arm failing the same clause. A consolidation lever lifted the best mined confidence from 0.533 to 0.872 on *unchanged evidence volume*, exposing bag churn — not evidence starvation — as the confidence ceiling's second cause.

The puzzle venue never produced a task score, and we claim none. What it produced is rarer in this literature: a negative whose root cause (vocabulary and seam structure, not mining capacity) was located, re-run as a controlled experiment in a sibling domain (110,701 band-shaped chain derivations where the old vocabulary produced zero), and folded forward as a **stamp-time check** — a venue whose declared percept vocabulary cannot feed the miner is now refused before any episode runs.

5 Why this works, and where it stops

Theory. The ledger buys three things search and scale do not. (i) *Calibration*: on controlled grid worlds, every rule the frequency channel could score read exactly its true frequency (mean $|\Delta f| = 0.0$, every seed) — the agent's confidence numbers mean what they say, so thresholds and gates are semantics, not decoration. (ii) *Attribution*: every conclusion carries its evidential base; when the adopted matrix in our MiniGrid domain inverted its own PASS under its control arm, the correction cost one query. (iii) *Compositionality of provenance*: because beliefs are objects, another system's verified beliefs can be imported, graded, and executed — the ARC-AGI-3 result is this property, measured.

Universality. Nothing in the pipeline is ARC-specific. The same mind-posture and harness have been carried to MiniGrid (policy-grade PASS, then honestly re-resolved), a third-party stochastic GridWorld (terminal documented negative), and MinAtar Freeway (stamp clean in one session) — four domains, one argument: the failure class that costs a campaign is now refused at stamp.

Limits, stated by us before the judges need to. The ARC-AGI-1 venue has no task score. The replay aggregate depends on the coverage of the recorded corpus. The clear chase proves the exploration frontier — not the rules — is the binding constraint on open-ended ARC-AGI-3 play, and our next lever (goal-shaped percepts; mined implications as a move generator for Thompson-sampled MCTS over the ledger's truth posteriors, rolled out on the replay harness) is specified but unmeasured. The agent is small, the worlds it has *mastered* are small, and we say so.

6 What this offers the 85% problem

The prize's question is not whether one team reaches 85% but whether anyone can. This submission contributes three transferable assets:

1. **A falsification-native loop for ARC-AGI-3.** Register predictions before outcomes; let falsifications revise rules, not narratives; gate between rounds on evidence rather than inside rounds on fear. Any solver can adopt this posture without adopting our engine.
2. **Artifact reuse as a first-class method.** Calibrate against a published corpus's own measurements, then execute its world models as rules. The 0.0 → 61.93 replay lift is a template for compounding the community's recordings instead of re-deriving them.
3. **The documented negative as an instrument.** Pre-registered gates, control arms as data, and verdicts whose numbers must resolve from the run directory they name. The fabricate-a-verdict failure mode — which our own adoption sessions exhibited three times and our audit rejected three times — is the exact post-hoc rationalization the field keeps diagnosing in LLMs. The fix is structural: make an unearned verdict *unrepresentable*, and it fails to parse into honesty.

7 Conclusion

The field's default response to ARC-AGI is to strengthen the answer machinery. We measured the alternative: an agent whose strength is its bookkeeping. It learned rules from raw experience with exact calibration, absorbed another system's recorded ARC-AGI-3 experience and executed it into real score, planned a 7-action clear from a model it never trained — and on the puzzle venue it published its refusal, root cause attached, and made the refusal permanent in its own pipeline. Reasoning, on this evidence, is not a bigger model. It is a ledger that the world is allowed to edit.

Reproducibility and provenance

- Engine: zMind (deterministic, seeded, single-threaded; byte-identical transcripts verified across independent processes). Wheels and commit SHAs pinned per run directory; version skew is caught by the parity suite, not by luck.
- ARC-AGI-1 venue: oracle, gate 0, lint, digest lock; round-1 report and round-2 documented negative retained with JSONL streams; corpus under the `arc-agi-v1` projection.

- ARC-AGI-3 work: calibration harness and report (`scripts/lp85_calibration/`, `runs/lp85-calibration/`), 25-game population run (`runs/awm-population/`), replay, cursor fix, hybrid explorer, and planning measurements in the Kaggle-side results record (61.93 aggregate / 132 levels / 11 wins / 7,740 actions; the 7-action planning proof over the artifact's own `move_map`).
- Reflection program: design and implementation plans, lever A/B records, gated pilot verdicts, live corpora with gate records, and the 5,000-action clear-chase corpus — all retained.
- Code and run artifacts are released under a permissive open license (CC0/MIT-0-class) per the competition's open-source requirement.

References

1. Chollet, F., et al. (2025–2026). *ARC-AGI-2 and ARC-AGI-3 benchmark materials*. ARC Prize Foundation.
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